Machine Learning Report: Sequential Data

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Report Task 5: Posterior Summary and $\hat{R}$ Interpretation

The following snippet shows the code I added in the Inference section, to generate the posterior summary and $\hat{R}$ values:

```
az.plot_trace(idata, var_names=["~mu", "~month mu_raw"])

# generate a tabular summary of posterior distributions together
# round the summary to 3 d.p.
az.summary(idata,var_names=["~mu", "~month mu_raw"], round_to=3)
```

The posterior distribution summary and the $\hat{R}$ values on pre-2020 data are shown in the table below:

	mean	sd	hdi_3%	hdi_97%	mcse_mean	mcse_sd	ess_bulk	ess_tail	r_hat
intercept	44559.890	1120.022	42663.783	46881.515	39.462	24.826	806.297	1404.106	1.001
temp coeff	-423.721	114.524	-637.080	-201.409	4.163	2.656	756.979	1286.382	1.001
linear trend	22.880	4.178	14.980	30.758	0.069	0.061	3624.182	2449.079	1.000
sigma	2658.697	147.176	2402.339	2944.790	2.365	2.617	3813.459	2817.412	1.001
month mu[January]	7514.545	889.977	5872.139	9185.640	26.026	15.174	1175.603	1776.563	1.000
month mu[February]	-573.155	868.706	-2247.725	996.039	23.271	13.513	1400.520	1939.042	1.000
month mu[March]	1451.421	760.346	24.717	2864.988	18.636	12.774	1671.096	2399.871	1.004
month mu[April]	-93.090	665.512	-1327.697	1170.224	10.731	9.889	3868.370	3139.195	1.001
month mu[May]	-780.078	690.399	-2094.315	481.149	13.213	10.849	2743.978	2429.293	1.001
month mu[June]	-1257.380	817.381	-2848.774	214.295	20.259	12.846	1624.829	2098.016	1.001
month mu[July]	-536.183	965.245	-2216.308	1422.127	28.412	17.787	1153.028	1905.189	1.000
month mu[August]	-2291.654	923.978	-4111.933	-671.937	26.095	16.213	1259.825	1862.120	1.002
month mu[September]	-2282.944	798.954	-3843.644	-857.079	19.480	12.673	1688.207	2054.130	1.000
month mu[October]	-306.046	679.347	-1493.619	1077.937	10.479	10.899	4216.896	2875.388	1.000
month mu[November]	-1479.813	739.710	-2833.064	2.911	16.398	11.365	2002.883	2580.397	1.001
month mu[December]	634.376	843.921	-959.011	2236.618	21.887	13.104	1484.939	2182.788	1.001

Figure 1: Tabular summary of posterior parameter estimates and $\hat{R}$ values.

Interpretation on $\hat{R}$ value

$\hat{R}$ compares variance within chains to variance between chains. If $\hat{R} \approx 1$, all the chains are sampling from the same stationary distribution, hence the MCMC is converged.

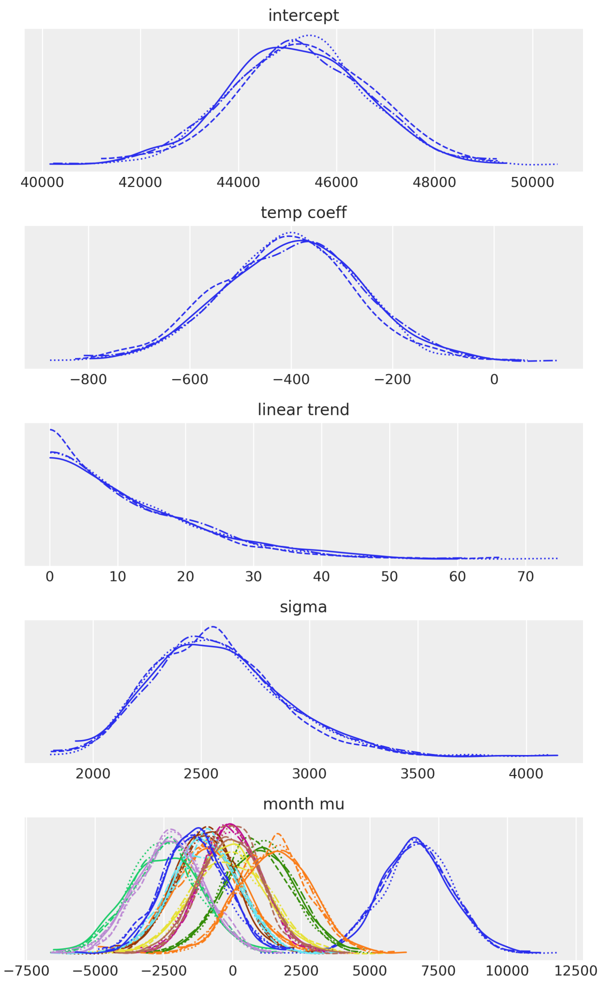
From the posterior summary (in Figure 1), all parameters including have $\hat{R}$ values extremely close to 1.00 . This indicates that the MCMC has converged.

Explanation of Poor Predictions in the Counterfactual Posterior Predictive Forecast

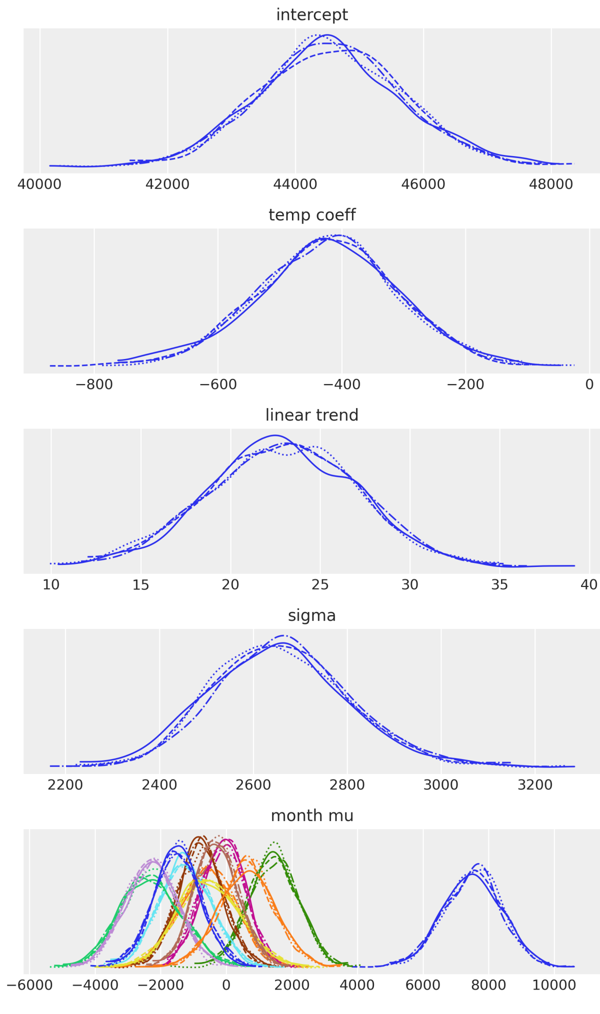
The model predicts poorly because it was trained on the pre-COVID death dataset. This model assumes that monthly mean death, temperature effects remains stable over time. When COVID–19 appeared, it introduced a structural break on these assumptions, and causing the counterfactual predictions to underestimate the reported death.

Report Task 6: Effect of Changing the Pre-COVID Training Period to 2010 on posterior distributions

Below is the comparison of the posterior distributions using pre-2010 and pre-2020 data.



Posterior distributions using pre-2010 data

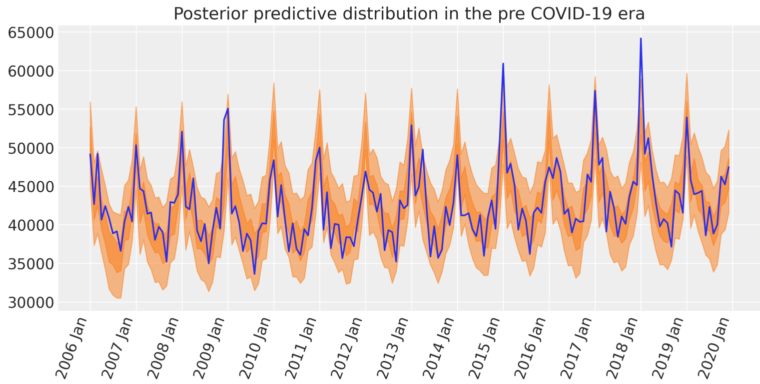


Posterior distributions using pre-2020 data

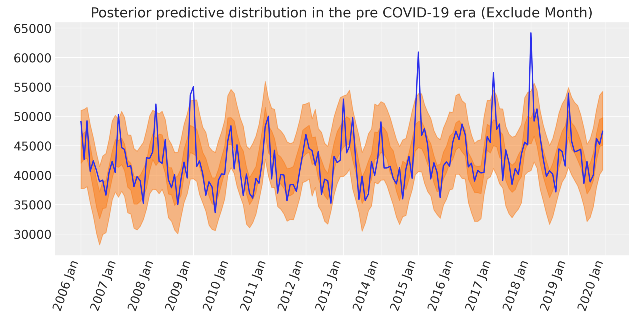
Figure 2: Comparison of posterior distributions between pre-2010 and pre-2020 data.

As shown in Figure 2, there is a noticeable difference on the linear trend. The pre-2010 linear trend is flattened and shifted towards negative compared to the pre-2020 one. This suggests seasonal and temperature effects remains similar across both periods, but the death trend is more stable before 2010 than in the period of 2010 to 2020 (i.e. the deaths gradually increased in 2010 to 2020).

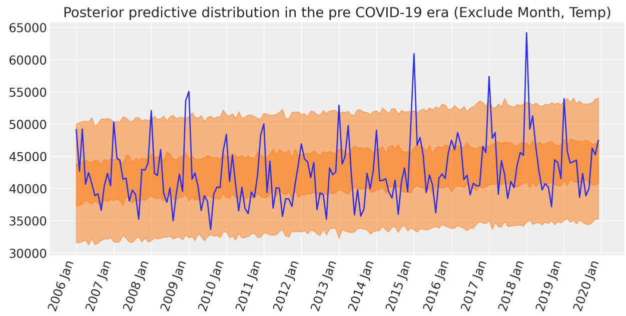
Report Task 7: Indicators Effect on Posterior Predictive Distribution



(a) Original model



(b) Without month indicator



(c) Without month and temperature indicators

Figure 3: Posterior predictive distributions under three model specifications for pre-2020 data

The original model (Figure 3a) has a narrow HDI band that fits to the reported deaths closely, indicating the model that captures the seasonal and temperature effect is able to predict accurate results on the training set.

The excluded month indicator model (Figure 3b) shows a wider range of the HDI band but it preserved the shape of the original model's HDI band. We can notice this model loses seasonal information on 2015 and 2018 spikes.

The excluded both month and temperature indicator model (Figure 3c) shows a very wide and flat shape HDI band. This suggests the model fails to capture the seasonal and temperature structure, leading to underfitting of the data.

Report Task 8: HMMs Comparison

Frequency plot of Supervised (HMM1) and Unsupervised (HMM2) sampled observations comparision

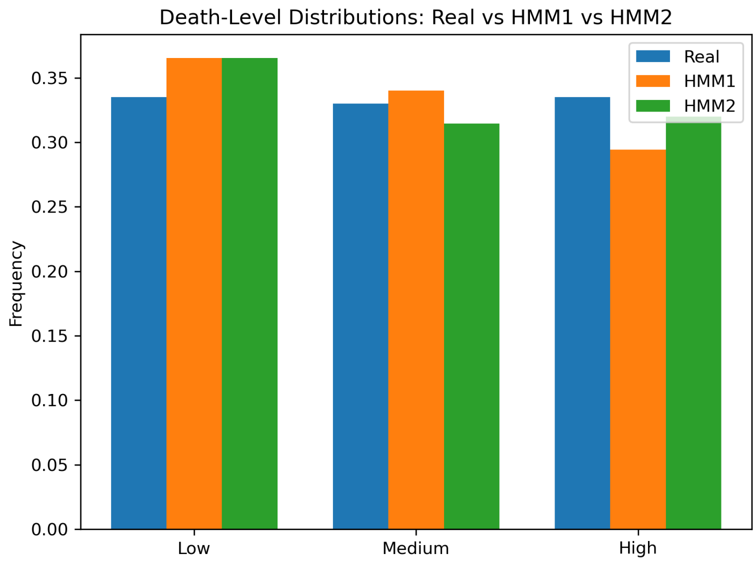
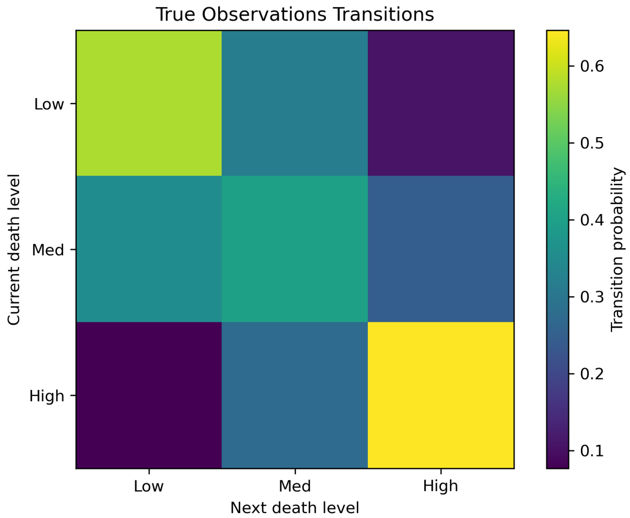


Figure 4: Frequency Plot of discretised death levels in the training data.

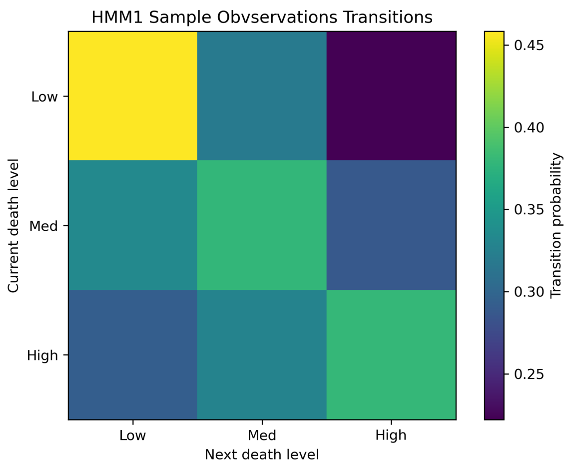
Figure 4 shows that both HMM1 and HMM2 produce marginal distributions that closely match the true observations, suggesting that both models capture the overall frequency of low, medium and high deaths reasonably well.

However, matching the marginal distribution alone is still insufficient, as the dataset also depends on the ordering (i.e., the timestamps) of the observations.

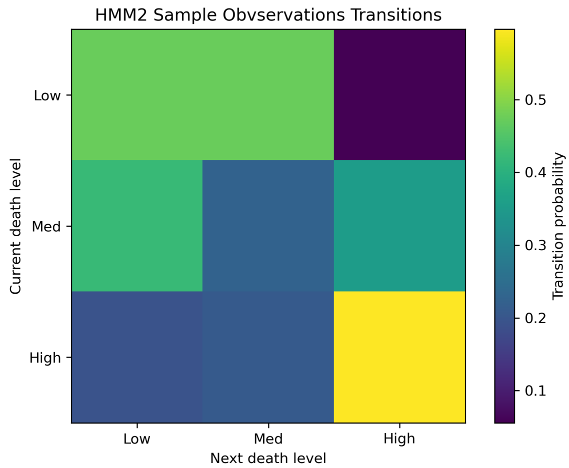
Comparison of Transition Matrices



(a) True Transition Matrix



(b) Sampled HMM1 Transition Matrix



(c) Sampled HMM2 Transition Matrix

Figure 5: Comparison of True Transition Matrix and sampled HMMs Transition Matrix

Figure 5, The HMM1 transition matrix matches the colour pattern of the true matrix more closely, or at least shows colours that are more similar to the unmatched regions in the True transition Matrix than the HMM2 transition Matrix.

This is expected, as the temperature state was provided as the latent variable for HMM1, allowing it to learn a transition structure much closer to the true sequence.